



Fig. 3: Circuit diagram of the project

correct location, open Arduino IDE. If you have pasted the folder in the correct location, you can find pde in File→Examples→pde. Open the code by proceeding as follows: File→Examples→pde→adioes

Connect the Arduino Uno board to your PC. From Device Manager, note the COM port at which the Arduino Uno board is connected. From Tools menu in the Arduino IDE, select the board as Arduino Uno, and the COM Port number noted earlier. Upload adioes code to the Arduino Uno board by pressing Upload button in the IDE.

Now copy the entire content of the extracted ArduinoIO folder to a folder in My Documents (in Windows PC).

This GUI application program has been developed in R2014a version of MATLAB. After installing this MATLAB version in your PC, open the file install_

arduino.m present in the directory where you copied contents of ArduinoIO folder. Now run the install_arduino.m file. This code will correctly install and save the path of the Arduino support package.

Open the source code file (Illumination001.m) for this project. Edit the line `a = arduino('COM9')` with the COM port number in your PC where the Arduino Uno board has been connected. When you run the project file, MATLAB will try to communicate with the board. After successful communication is established, you can control LEDs by pressing the corresponding pushbuttons in the GUI. **EFY**

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com



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Saikat Patra is passionate about electronics and MCU-based embedded system applications

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